

Demo: Out-Of-Body Experience induced after Virtual Embodiment

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Figure 1: First line: Visuomotor phase, second line: Visuotactile phase, third line: Out-of-Body phase

ABSTRACT

Out-of-body experiences (OBEs), where individuals perceive themselves as separated from their physical bodies, hold significant potential for research in psychology and neuroscience. We present a demonstration of a system designed to induce OBEs through immersive technology and virtual body ownership illusions (VBOI). Our approach uses personalized avatars, synchronized visuomotor and visuotactile stimulation, and real-time interaction to create strong VBOI, followed by the induction of OBEs via controlled camera movements. The system is lightweight, user-friendly, and offers better ecological validity, opening new avenues for studying mind-body relationships in controlled environments. Applications include advancing research on self-consciousness, embodiment, and disembodiment, as well as exploring therapeutic interventions for conditions such as PTSD and anxiety. In future research, we wish to explore OBEs in more realistic contexts, such as VR car crash scenarios.

Index terms: Virtual Reality, Body Ownership Illusion, Out-of-Body.

1 INTRODUCTION

Out-of-body experiences (OBEs) can be described as phenomenon in which a person appears to be conscious and perceives their body and surroundings from a position located outside their physical body [1]. In other terms, an OBE is an event

in which you perceive yourself as being separated from your physical body, being able to see, feel, and move around without it [2]. OBEs are gaining importance for research in psychology and neuroscience, which recognize the experience of embodiment as fundamental in self consciousness [3].

Recent advances have enabled the induction of OBEs through virtual reality (VR) technology in controlled environments, allowing researchers to simulate the sensation of perceiving one's body from an external perspective [4]. To achieve a strong OBE illusion, a strong ownership illusion toward the virtual body is required in a previous phase [5]. Indeed, if participants do not experience a sense of being "in" the virtual body (VB), they would very unlikely experience being "out" of it. Even when the viewpoint transitions from the avatar's eye position to a location distant from the body, such as above the avatar or near the ceiling.

Although the induction of OBE illusions is not a novel concept—having been previously achieved using real-time video [6] or virtual reality with avatar embodiment [7]—advances in technology have enabled the development of lighter, more comfortable, and less cumbersome setups, hence enhancing ecological validity. Our demonstration builds on this progress, utilizing virtual reality technology to induce OBEs with the aim of enhancing users' awareness of their embodied self. We believe this approach can introduce new opportunities for both research and therapeutic applications.

2 MOTIVATION AND RESEARCH GOALS

The main motivation of this work is to demonstrate that using immersive VR with avatar and body ownership illusion, it is possible to induce complex illusions such as OBEs, under relatively easy conditions, not requiring a laboratory.

Our research goals include:

- **Exploring Virtual Body Ownership Illusion and OBE Simulation.**
- **Exploring the Mind-Body relationships through virtual body illusions.**
- **Providing a Tool for Neurosciences and Psychology studies.**

Beyond demonstrating the ease of use of our system, we seek to understand whether OBE simulations could help individuals with mind-body-related issues, and more generally study the connection between the body, the mind and the physical self.

3 SYSTEM OVERVIEW AND ARCHITECTURE

Based on Avaturn [8] and QuickVR [9] our system combines immersive VR technology with synchronous real-time avatar to create, in a first phase, (Figure 1, Line 1 and 2) a highly interactive and personalized virtual body ownership experience. In a second phase (Figure 1, Line 3), an OBE is induced by a slow camera movement.

Key components include:

- **Virtual Reality Environment:** Users wear a commercial 6 degree-of-freedom VR headset displaying an immersive interactive environment. Additionally, they can interact with the virtual environment using the hand controllers.
- **Body Representation:** The system uses realistic avatars which can be created from participants' photographs. These avatars are rendered in real-time within the environment. Participants can control the movements of their virtual body and interact with the virtual environment. This is designed to enhance the body ownership illusion toward the virtual body.
- **Visuomotor and visuotactile stimulation:** Our system allows virtual reality simulations with visuomotor synchrony. For example, moving in front of a mirror and visualizing in realtime the results of the movements in the mirror or looking at the avatar, or grabbing and wearing a cap, seeing the avatar putting the cap on his head. It also allows visuotactile simulation, generating vibration in the hand controllers in response to an event, like when tracing the lines of drawing or when touching balloons to make them pop.
- **Out-of-Body Experience simulation:** During the last visuotactile stimulation, while the participant is busy popping balloons, the camera starts slowly to move away from the virtual body. The camera exits the virtual body and gradually rises towards the ceiling. After reaching the final height, the point of view remains stable for a moment, during which the participant can observe the avatar sitting on the chair below.
- **Head and hands movement monitoring:** Real-time monitoring of physical metrics, such as the movement of the hands and the head, are recorded during the whole experience by the system. This allows to track and analyse the body's response during the experience and provides data on how users react within the OBE simulation.
- **Questionnaire responses:** Questionnaires can be displayed and answered by participants while wearing the head-mounted-display and experiencing the illusion in the virtual environment.

4 APPLICATIONS AND NOVELTY

This system presents several unique and innovative features:

- **Enhanced Embodiment:** In the first phase, our system aim at inducing a strong virtual body ownership illusion by combining synchronous multisensory stimulations. We believe this could help define validated protocols for inducing strong virtual body embodiment.
- **Psychological Research:** The system offers a controlled environment for studying virtual body ownership, embodiment, self-perception, self-consciousness as well as disembodiment and consciousness "out-of-the-self". Researchers can test hypotheses about "embodiment" and "disembodiment" or study how detachment and reconnection influence self-awareness, body ownership. More over the system could also be applied to study pain perception, time perception, memory recording or memory retrieving and other sensory experiences, in relation with self-location.
- **Therapeutic Potential:** The OBE simulation could serve as a therapeutic tool for mental health conditions, including PTSD, anxiety, or dissociation. By giving patients a safe, controlled way to disconnect from their body, the system may help them reframe and regain control over their embodied experiences, especially in the context of trauma recovery.

5 FUTURE RESEARCH DIRECTIONS

Further than demonstrating the ease of use and efficiency of our system, we wish to collect feedback to improve it and continue working on studies related to OBE. In particular, we aim to study OBE in more realistic context, like car accident situation where after suffering a moderate car-crash situation, participants undergo an OBE. This would allow us to study these OBEs and possibly compare them with those happening in real-life situations.

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VIDEO OF THE DEMO

<https://youtu.be/wWhwkRvF9ps>

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